

Blood Simple



“Decoding your blood drop by drop”

A webapp to help you understand your blood sample

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Where are we ?

Text extraction from PDF and image

```
$ py data_loading.py
```

From PDF :

```
('RBC Count ', 5.0, 14.0, 16.0, 'g%')
('PCV ', 36.0, 35.0, 45.0, '%')
('MCV ', 72.0, 80.0, 99.0, 'fl')
('MCH ', 30.0, 28.0, 32.0, 'pg')
('MCHC ', 41.67, 30.0, 34.0, '%')
('RDW ', 10.0, 9.0, 17.0, 'fl')
('Total WBC Count ', 5500.0, 4000.0, 11000.0, '/ cu.mm')
('Neutrophils ', 60.0, 40.0, 75.0, '%')
('Lymphocytes ', 30.0, 20.0, 45.0, '%')
('Eosinophils ', 5.0, 0.0, 6.0, '%')
('Monocytes ', 5.0, 0.0, 10.0, '%')
('Basophils ', 0.0, 0.0, 1.0, '%')
('Platelet Count ', 1550000.0, 150000.0, 450000.0, '/ cu.mm')
```

From Image :

```
'RBC Count ', 5.0, 14.0, 169.0, '%')
'PCV ', 36.0, 35.0, 45.0, '%'
'MCV ', 72.0, 80.0, 99.0, 'fl')
'MCH ', 30.0, 28.0, 32.0, 'pg')
'MCHC ', 41.67, 30.0, 34.0, '%'
'RDW ', 10.0, 9.0, 17.0, 'fl')
'Total WBC Count ', 5500.0, 4000.0, 11000.0, '/ cu.mm')
'Neutrophils ', 60.0, 40.0, 75.0, '%'
'Lymphocytes ', 30.0, 20.0, 45.0, '%'
'Eosinophils ', 5.0, 0.0, 6.0, '%'
'Monocytes ', 5.0, 0.0, 10.0, '%'
'Basophils ', 0.0, 0.0, 1.0, '%'
'Platelet Count ', 1550000.0, 150000.0, 450000.0, '/ cu.mm')
```

Metrics :

PDF Extraction Time: 0.04597067832946777 seconds
Number of Lines Extracted from PDF: 13
Image Extraction Time: 2.873901605606079 seconds
Number of Lines Extracted from Image: 13
Number of Common Lines: 12

	# 004, Wap-2, New Plaza, Post Nagar, Dindur (S) Mumbai - 400 075	E-mail: contact@crystallabinc.com crystallabinc@gmail.com www.crystallabinc.com	
Crystal Data Inc. CONSULTING, DEVELOPMENT, SUPPORT		Phone: 022-21022430 / 21010491 / 02223408 Mobile: 9302727090 / 9302340018	ISO 9001:2015
LAB NO.	S	DATE	10-Aug-2011
PATIENT NAME	MS. KETAN CHAVAN	SEX	Male
REF. BY DR.	DR. PATIL M.B.B.S.	AGE	29 Years
SAMPLE COLL. AT	CRYSTAL LAB		
COMPLETE BLOOD COUNT			
TEST	RESULT	REFERENCE RANGE	
Hemoglobin	15	Male: 14 - 16 g/dl Female: 12 - 14 g/dl	
RBC Count	5	3.0 - 4.4 %	
PCV	36	30 - 44 %	
MCV	72.00	80 - 100 f	
MCH	30.00	28 - 32 pg	
MCHC	41.67	30 - 34 %	
RDW	10	9 - 17 f	
TOTAL WBC COUNT			
Total WBC Count	5500	4000 - 11000 / cu.mm	
Neutrophils	60	40 - 75 %	
Lymphocytes	30	20 - 45 %	
Eosinophils	5	00 - 06 %	
Monocytes	5	00 - 10 %	
Basophils	0	00 - 01 %	
PLATELETS			
Platelet Count	1550000	150000 - 450000 / cu.mm	
Platelets on Smear	Adequate On Smear		
PERIPHERAL BLOOD SMEAR			
RBC Morphology	Normocytic, Normochromic		
WBC on PG	Normal		
Test done on Hitachi Kubota BCV-400P fully automated cell counter			
----- End of Report -----			
 Chander Varak D.M.L.T. Hospitalized Patient (K) and Indicate Abnormal Report Forward by Mail to your physician only		 Dr. Parag Shah M.D. M.B.B.S.	

Database creation with reliable sources

2 tests : complete blood count (CBC), comprehensive metabolic panel (CMP).

29 parameters.

77 diseases linked to the parameters and their prevalence/incidence.

1st version of the webapp using Flask

A first working demo :

<http://127.0.0.1:5000>

Project Assessment

'Easy' steps

- Getting started on the project
- Finding blood sample results
- OCR / PDF extraction

Challenges faced

- Finding reliable sources for habit/diet advices and sometimes for epidemiology
- Designing a mobile app (abandoned)
- Designing a webapp on Django (eventually abandoned to Flask)
- Integrating text extraction

Where are we going?

Incoming tasks

- Finishing the database
- **Integrating properly the various features in the webapp**
- Appearance (conditional formatting? visual identity?)
- New features : FAQ or chatbot using LLM, user input about age/gender, test compatibility...

Thank you for listening

Questions?